

Education

Indian Institute of Technology (IIT) Goa	In Progress	M.Tech in VLSI and Signal Processing	2024 – Present
Sinhgad Institute of Technology, Lonavala	CGPA: 7.64	B.E. Electronics & Telecommunication	2017 – 2021

Academic Research

Master's Thesis: Smart Periodontal Probe Design with Embedded Sensors <i>Guide: Dr. Bidhan Pramanick, IIT Goa</i>	2024–Present
- Designing smart periodontal probe with integrated biomedical sensors. - Implemented signal processing for accurate gum depth/tissue response. - Modeled in COMSOL; focus on low-power and ergonomic design.	

Teaching Assistantship

Teaching Assistant: Data Communication Lab <i>Guide: Dr. Neelakandan R, IIT Goa</i>	Jul 2025 – Present
- Guided 40+ students in data communication and error detection experiments - Reduced experiment downtime by 25% via troubleshooting support. - Streamlined lab operations with proactive equipment setup.	

Internships

Full-Stack Data Engineering Intern - Built predictive data models & sentiment analysis pipelines in Python, improving accuracy by 5 - Led 2-member team to streamline deployment process. [Certificate]	Expertshub, Allahabad, 2021
Cloud & Web Development Intern - Developed web applications and improved UI design using HTML5 - Built and deployed scalable cloud-based services on AWS using Python, reducing deployment overhead by 10 - Delivered a comprehensive final report with full documentation [Certificate]	Techmaniacs EduServices, Lonavala, India, 2021
IoT Intern - Designed IoT applications integrating multiple sensors and microcontrollers for real-time data collection. - Engineered a Google Assistant-controlled robot, enabling seamless voice-to-action execution via IFTTT. - Authored detailed documentation, improving reproducibility and project handover efficiency. [Certificate]	Expertshub, Bangalore, India, 2021

Projects

Two-Stage Operational Amplifier (Cadence) <i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2025
Designed and simulated a two-stage op-amp (65nm PDK) in Cadence Virtuoso; validated with DC, AC, and transient analyses [Report].	
4-bit Adder Design & Delay Analysis (Electric VLSI) <i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2025
- Implemented a ripple-carry adder; verified functionality and timing in LTspice. - Analyzed propagation delays; critical path delay = 547.32 ps [Report].	
Energy Management System (FSM, VHDL) <i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2024
Developed a finite-state machine for dynamic energy load management using Quartus and ModelSim [Paper].	
NMOS Differential Amplifier (Electric VLSI) <i>Guide: Dr. Nandakumar Nambath, IIT Goa</i>	2024
Implemented and analyzed an NMOS differential amplifier; measured gain and output swing in LTspice [Report].	
Audio Equalizer (DSP Kit) <i>Guide: Dr. Sujit Kumar Sahoo</i>	2024
Built a real-time Chebyshev filter-based audio equalizer on a TI DSP kit; tested using Keil µVision5.[Code]	

IoT-based Health Monitoring Kit (Final Year Project)

2021

- Developed an IoT-based kit integrating vital-sign sensors and cloud monitoring; showcased at TECHNOVATION 2K21. [Presentation], [Video], [Certificate].

Additional Projects

Drone Project (2018)

SIT, Lonavala

- Engineered Arduino Uno-based quadcopter for land tracking and mapping.
- Programmed predefined flight trajectories for stable navigation. [Photo].

Virtual Assistant "Christopher" (2019)

SIT, Lonavala

- Developed Linux-based virtual assistant for routine task automation. [Video Clip].

Volunteer Experience

Speaker, IIT Goa – Conducted PCB Design & Soldering Workshop (Eagle). [Certificate] 2025

Seminar Mentor, IIT Goa – Inspired PM Shri KV students with drones & embedded demos [Photo] 2025

IoT Mentor – Guided Smart Mirror project (SIT, 2020) [Certificate] 2020

Skills

Programming	VHDL, Verilog, Python, C/C++, Java, MATLAB
Tools	Cadence, LTspice, Electric VLSI, Quartus, Arduino IDE
Domains	VLSI Design, DSP, Embedded Systems, Biomedical Sensors, IoT Healthcare
Cloud	AWS, Google Cloud, Intel OpenVINO Toolkit

Certifications

- Advanced Drone Technology (Air Taxi) Workshop – Bharat Space Education Research Centre & I-STEM, hosted at IIT Goa, 2025 [Certificate]
- MATLAB Onramp (Mathworks, 2021) [Certificate]
- Google Cloud Certified Professional Cloud Architect (Coursera, 2020) [Certificate]
- Neural Network from Scratch in TensorFlow (Coursera, 2020) [Certificate]
- Intel OpenVINO Toolkit for Computer Vision (Coursera, 2020) [Certificate]
- Fundamentals of Parallelism on Intel Architecture (Coursera, 2020) [Certificate]
- ISRO Online Learning – "Application of Geoinformatics in Ecological Studies" (ISRO, 2020) [Certificate]
- Microprocessor & Microcontroller Certifications (Bharat Acharya Education): 8085 (2023) [Certificate] | ARM (2020) [Certificate] | 8051 (2020) [Certificate] | 8086 (2020) [Certificate] | COA (2020) [Certificate]

Achievements

Model United Nations	Represented Panama at BITS Goa MUN (UNSC), debated arms trade & counter terrorism [Certificate]	2025
GATE EC	Scored 340/1000, All India Rank (AIR) 9467 [Scorecard]	2024
TCS iON NQT	Cognitive: 1101.32/1800 (61.18%) and Psychometric Assessment [Scorecard] [Report]	2023
TECHNOVATION 2K21	State Level Project Presentation [Certificate]	2021
Elocution Competition, SIT Lonavala	1 st Prize [Certificate]	2021
Model Mania Exhibition	Won against all SIT colleges; 1 st Prize, Sinhgad Tech Fest, Prize: 10,000 [Certificate]	2019
Bytes of India Feature	Drone Project showcased and recognized [Article]	2019

Positions of Responsibility

PG General Secretary, IIT Goa	Elected liaison for PG students, 2025–26	2025
Core Team Member, Karandak Event, Sinhgad Institute of Technology	Organized 12 inter-college events with a 10-member team; increased participation 15% [Certificate]	2020